

Data Driven Education

NLP Subteam

Final Presentation

A dark blue diagonal gradient bar that starts from the bottom left and extends towards the top right, covering the lower half of the slide.

Goals

- *Lay strong foundations with progress that helps future sub teams scale the application*
- *Ensure that the application is scalable beyond OMSA and benefits all undergraduate/graduate courses at GT and beyond*
- *Develop tools to help instructors understand students better through real-time reports which ultimately resulting in improving student academic performance*

Semester Overview

Phase 1: August 28 - September 20

- *Understanding the problem*
- *Researching Ideas*
- *Potential Solutions*
- *Researching pros and cons of the solutions*
- *Understanding challenges*

We met a total of 2 times during this period: Meeting minutes are posted at

[09/09/2019 Meeting](#)

[09/20/2019 Meeting](#)

Semester Overview

Phase 2: September 20 – October 17

- *Collecting additional data (5623 posts) from Fall19 and Spring 2019 ISYE 6501 classes*
- *Getting familiar with NLP scripts and data provided by Dr. Staudaher*
- *Extracting data from JSON file*
- *Pre-processed the post contents*
- *Perform basic sentiment analysis using textblob module*

We met a total of 3 time during this period: Meeting minutes are posted at

[09/27/2019 Meeting](#)

[10/04/2019 Meeting](#)

[10/17/2019 Meeting](#)

Semester Overview

Phase 3: October 17 – November 15

- *Data visualization: Demonstrating sentiment statistics through Jupyter Notebook*
- *Discussion on scaling the project*
- *Research on Web application*
- *Setting up the environment*
- *Web Application Mockup*
- *Documented current and future plans for web application*

We met a total of 3 time during this period: Meeting minutes are posted at

[11/01/2019 Meeting](#)

[10/08/2019 Meeting](#)

[10/15/2019 Meeting](#)

Data Description

Summary Statistics

	Total Posts	Active Users
Course		
ISYE- 6501-OAN Fall 2018	1701	320
Verified MM Learners ISYE- 6501 Fall 2018	448	118
ISYE- 6501-OAN Spring 2019	2550	467
VERIFIED MM LEARNERS ISYE- 6501 Spring 2019	863	189
Total	5562	1094

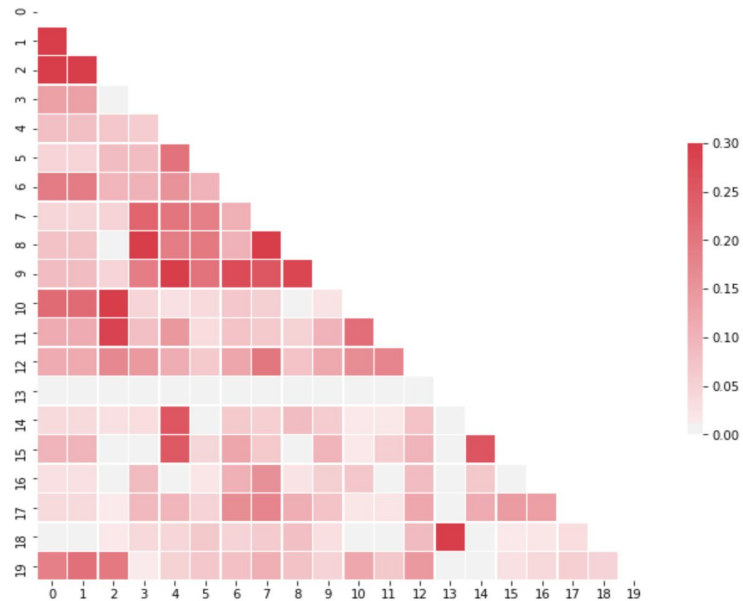
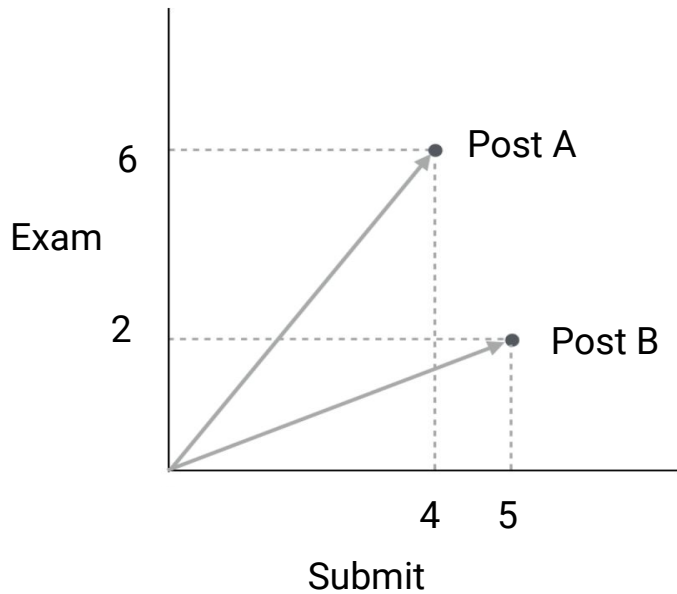
Word Cloud of Most Frequent Words



Data Pre-processing

- *Tokenizing*
- *Stemming*
- *Stop-words Removing*
- *Feature Extraction (TF-IDF)*

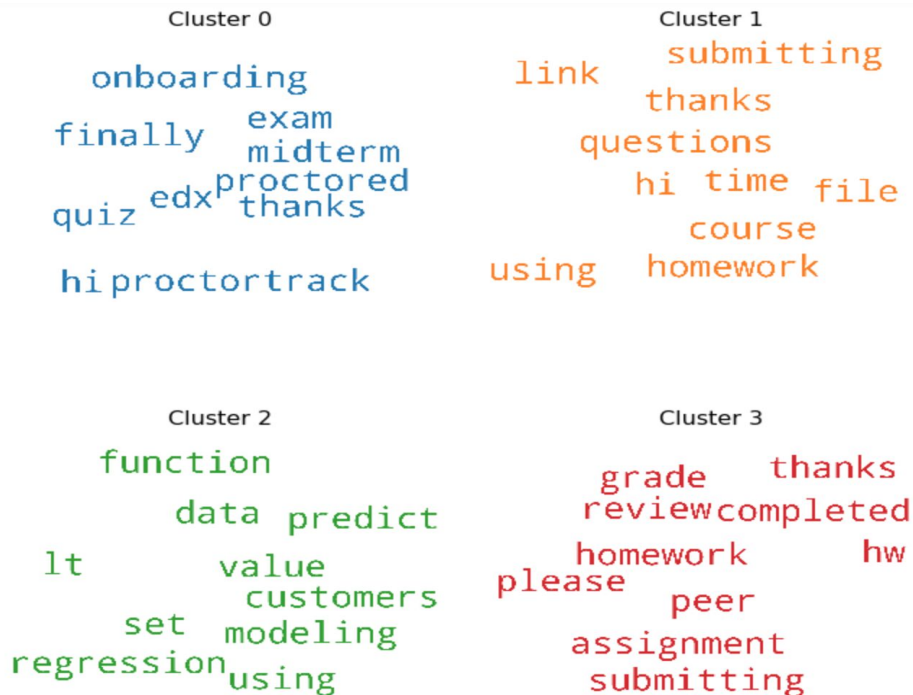
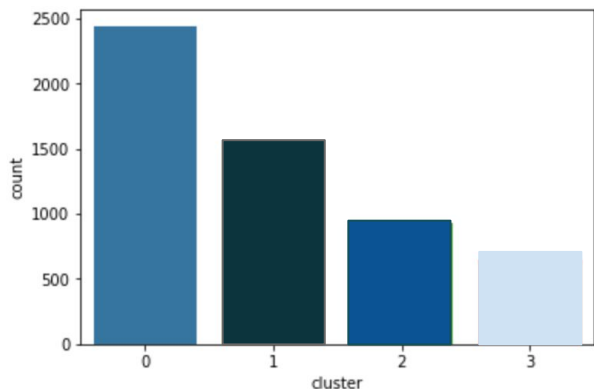
Posts Similarity



Clustering (K-means)

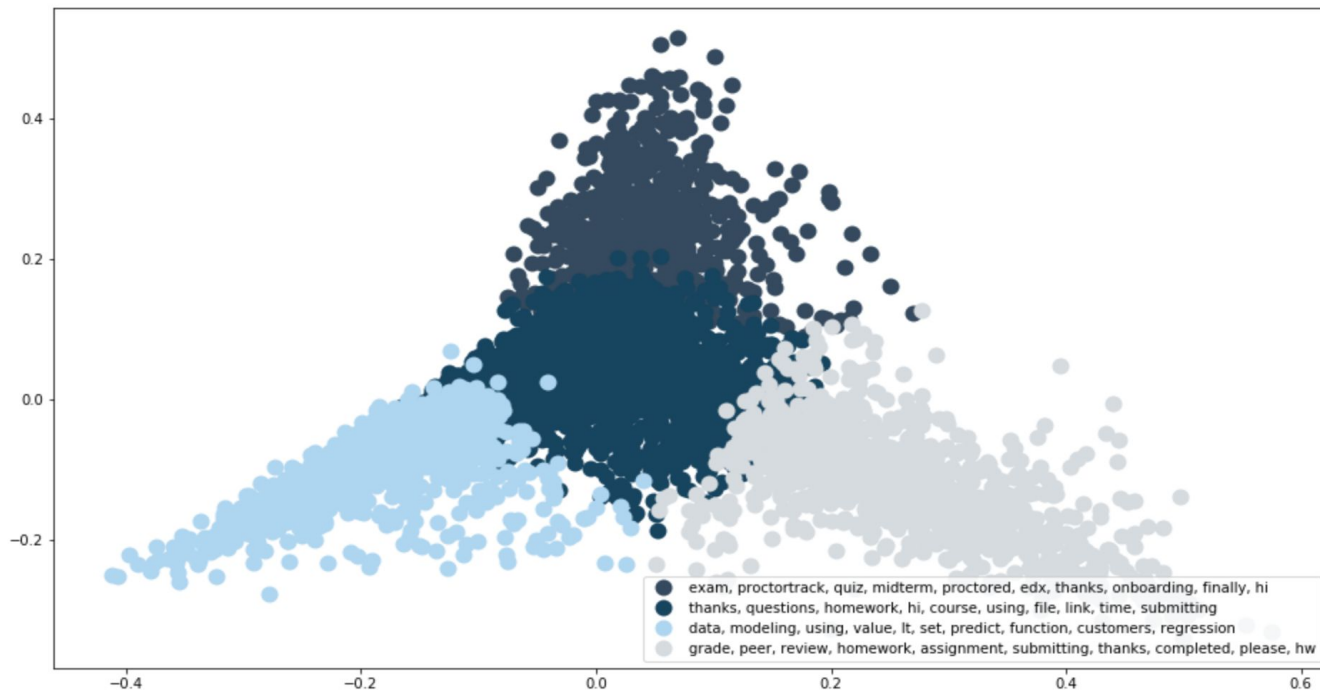
Word Clouds of Top 10 Keywords in Each Cluster

Number of Posts by Cluster



K-means Clustering

2D Visualization of Results



Topic Modeling – LDA

Number of Posts by Dominant Topic

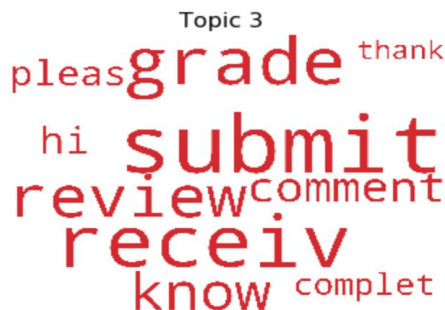
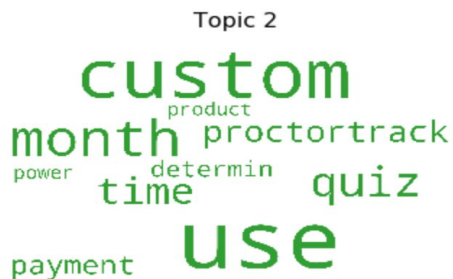
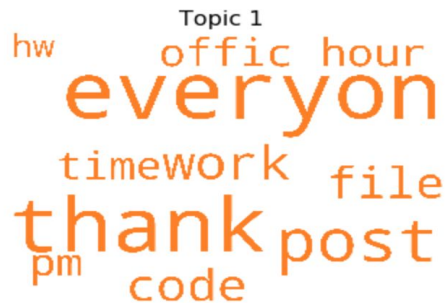
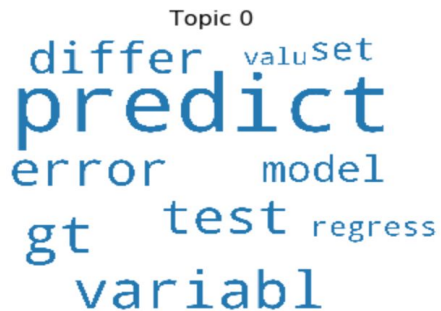
topic	
0	1987
1	1603
3	1215
2	757

Dominant Topic

	Topic0	Topic1	Topic2	Topic3	topic
Post0	0.01	0.96	0.01	0.01	1
Post1	0.00	0.00	0.00	0.99	3
Post2	0.00	0.00	0.00	0.99	3
Post3	0.01	0.25	0.01	0.74	3
Post4	0.01	0.17	0.01	0.81	3
Post5	0.00	0.45	0.46	0.09	2
Post6	0.30	0.14	0.56	0.00	2
Post7	0.01	0.43	0.01	0.55	3
Post8	0.01	0.98	0.01	0.01	1
Post9	0.20	0.44	0.01	0.35	1

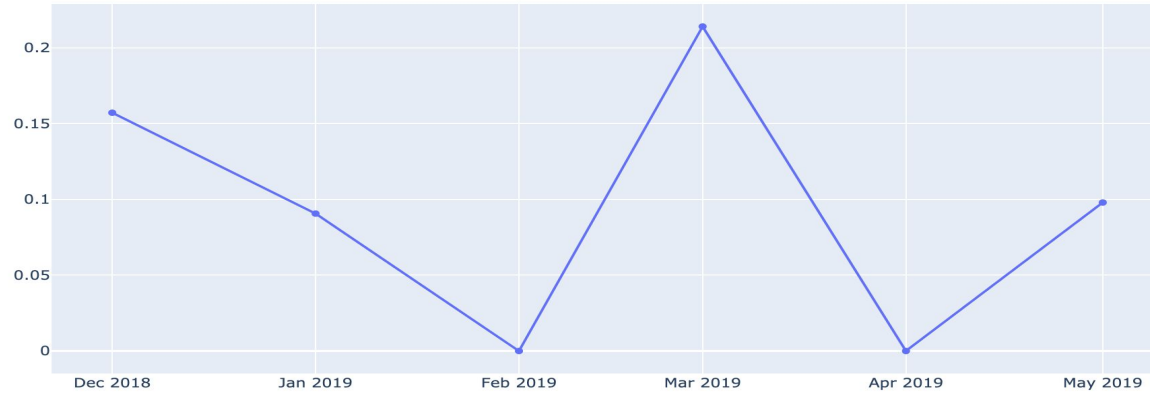
LDA

Word Clouds of Top 10 Keywords in Each Topic



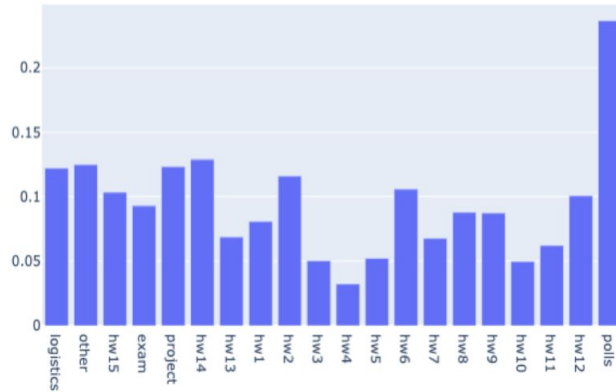
Sentiment Analysis

Sentiment Over Months



- Some other visualizations we created over the semester
- Still working on flagging low-sentiment categories and adding other features

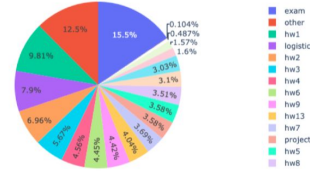
Average Sentiment by Category



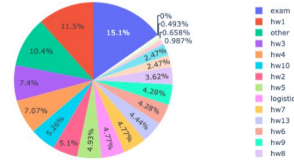
Visualizations with Plot.ly

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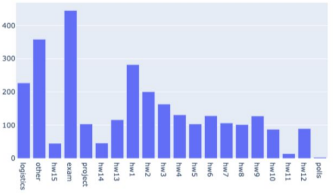
Post Content Distribution by Category



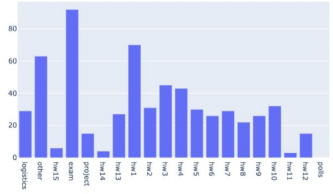
Negative Post Content Distribution by Category



Post Content Distribution by Category

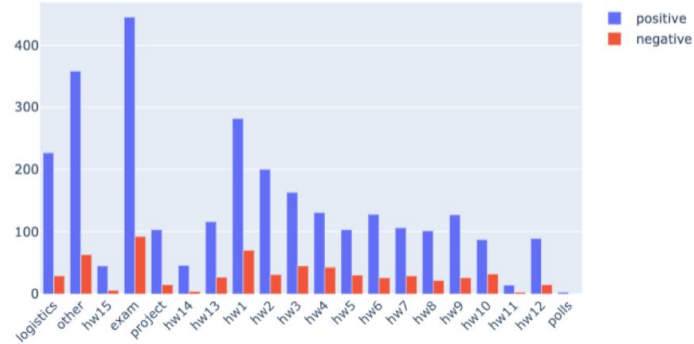


Negative Post Content Distribution by Category

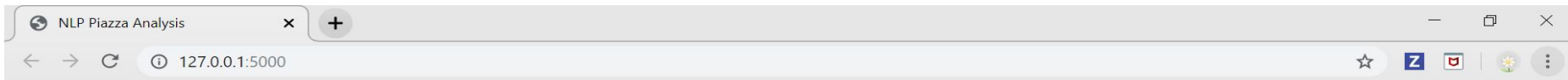


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Post Content Distribution by Category



The Goal: Web App



Please input your piazza credentials

Username:

Password:

Web App

- Prototyped web application
- Documentation of current application
- Outline of tasks to accomplish as well as ideas of how to implement

Documentation

Trello

- *Our project phases, meeting minutes(Google drive), and tasks*

[Link to our Trello Board](#)

Github

- *Notebooks, presentations, codes*
- *Notebooks are under “notebooks” folder inside C21U /vip-nlp*
- *Presentation are under “presentations” folder inside C21U /vip-nlp*

All code commits are under “nlp_example” and “src” folder with appropriate documentation

Please contact Dr. Staudaher for GitHub access

Semester Challenges

- *Getting familiar with the project (New sub-team with 4 new members)*
- *Limited data set*
- *Time commitment vs. ideal solution*
- *Limited experience*
- *Web App*

Advice to future sub-teams

- *Get in touch with Dr. Staudaher ASAP*
- *Analyze sub-teams*
- *Connect your passion and strength with the sub-teams*
- *Writing software for production sounds good but time constraints makes it challenging*
- *Prior research and preparation is helpful*
- *Don't be afraid to ask question (We also felt lost at the beginning)*
- *Lots of opportunities!!!*

Big Thanks to *Dr. Staudaher* and all the advisors!!

Good luck for your Finals!